

In the MIC of time to fight antimicrobial resistance

Antimicrobial resistance is an escalating global threat that can have disastrous consequences during the treatment of severe cases of infection. Accurate and rapid AST systems are required in the fight against resistance and to save lives. This is where **Q-linea's** rapid AST system, ASTar, can help save lives.

Bloodstream infections (BSIs) are a major cause of global mortality, morbidity and healthcare costs. Without swift and appropriate antibiotic treatments, BSIs can progress into sepsis, a life-threatening condition of organ dysfunction caused by a dysregulated host response to infection.

Sepsis affects nearly 50 million people around the world every year, where, tragically, 11 million people lose their lives to sepsis or sepsis-related complications. The landscape of sepsis is treacherous and challenging – marked by a pathophysiological cascade of events triggered by the immune system that creates a complex medical condition that must be treated in an apt and timely manner.

The initial standard of care for BSI and septic patients is to treat with broad brushstrokes using a palette of antimicrobial agents that are oftentimes suboptimal or ineffective. While this initial empiric therapy is necessary prior to pathogen identification and tailoring of treatment, it has a significant impact on patient well-being and contributes to the rising emergence of antimicrobial resistance (AMR).

Deaths directly related to AMR are projected to rise beyond ten million by 2050, predominantly due to antibiotic misuse.

As resistance continues to increase on a global scale, the armoury of effective antimicrobials diminishes, meaning AMR demands attention now, while we can still act. It is a significant concern for the continued effectiveness of antibiotics used to treat severe infections.

Expedited AST using ASTar

Rapid diagnostics and effective treatments are paramount for reducing mortality and saving lives in time-sensitive cases of severe infection, such as sepsis.

Time to optimal treatment remains a major contributing factor when managing



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sepsis patients, where every hour of delay in appropriate antimicrobial treatment detrimentally impacts clinical outcomes. Prolonged inappropriate therapies also contribute significantly to the emergence of AMR in the broader community.

Adjustment of empiric therapies into optimal antimicrobial treatments can take several days when using conventional antimicrobial susceptibility testing (AST) methods. With Q-linea's rapid AST system, ASTar, minimum inhibitory concentrations (MICs) and susceptibility categorisation (SIR) are delivered in around six hours, significantly reducing the time it takes to begin tailoring a patient's antimicrobial therapy, sometimes up to 48 hours faster than conventional methods.

The fully automated ASTar system shortens the clinical workflow and enables predictable turnaround times, where testing can begin directly from a positive blood culture sample independent of pathogen ID, and a comprehensive AST report is delivered within the same work shift.

This benefits not only the patient's clinical course but also alleviates many of the challenges burdening modern healthcare.

Faster treatment optimisation would be expected to reduce the length of ICU and hospital stay, as well as decrease the duration of antibiotic course, in turn supporting a significant decrease in healthcare costs.

This has been backed by several rapid AST performance studies, which Q-linea hopes to replicate in its ongoing lifetimes health economic and outcome research (HEOR) where it has implemented ASTar in multiple hospitals across Italy. The study aims to investigate the impact of introducing rapid AST into the diagnostic workflow and the guidance it can offer in regions of high AMR where effectual treatment revision is not always apparent.

Unlike other AST systems that extrapolate MIC results from a limited number of antimicrobial dilutions, ASTar utilises a broth microdilution-based technology with a controlled inoculum, thereby mitigating the risk of the inoculum effect.

ASTar supports a broad antimicrobial panel across wide dilution ranges to generate accurate and true MIC results, helping physicians select the best antimicrobials and dosages in just a single



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AST run, without the need for follow-up testing. Earlier initiation of optimal therapy is crucial for better long-term outcomes. Survivors often grapple with a life-altering aftermath that could be mitigated by more responsive interventions that reduce the risk of morbidities associated with suboptimal therapies and long hospital stays.

ASTar publications

In an era where rising rates of AMR are making it increasingly more complicated to treat patients successfully, Q-linea believes ASTar is well equipped to aid in the management of BSIs in even the most challenging cases. This has been reflected in several ASTar performance studies.

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In fact, Q-linea was proud to be acknowledged for having the best clinical microbiology publication with respect to rapid AST during ASM 2023. The paper ‘Performance of a System for Rapid Phenotypic Antimicrobial Susceptibility Testing of Gram-Negative Bacteria Directly from Positive Blood Culture Bottles’, was published in the Journal of Clinical Microbiology and summarises the results of the ASTar European clinical performance

study. Here, Q-linea demonstrated the high accuracy and consistent reproducibility of ASTar when compared with broth microdilution (BMD) for a wide range of fastidious and non-fastidious bacteria and a broad range of antibiotics including ceftazoxime/tazobactam and ceftazidime/avibactam, as well as colistin, a ‘last-resort’ antibiotic. Q-linea was pleased to prove that ASTar is already well prepared to empower clinicians to optimally manage BSIs caused by a breadth of pathogens, facilitated by a broad drug panel that has the potential for future expansion with novel antibiotics.

An evaluation conducted in Whiston, UK, showed the enormous difference ASTar can make during routine testing when

implemented in a 24/7 microbiology laboratory, especially in cases of resistant infections. By reducing time to optimal treatment by over 24 hours when compared with the current standard of care, ASTar prompts faster antimicrobial adjustments when time matters most and would ensure patients are more quickly placed on regimens that truly benefit them.

Additionally, Q-linea carried out a retrospective study in Sevilla, Spain, a region

with high prevalence of AMR, and showed that ASTar had a strong overall agreement with reference BMD methods in *E. coli* isolates harbouring *bla*_{OXA-1} and *bla*_{TEM-1}. Resistance genes associated with poor reliability in piperacillin-tazobactam (PTZ) AST from commercial panels that frequently yield MIC values within the area of technical uncertainty category (ATU), possibly contributing to increased mortality.

Because of this, several guidelines have recently rejected the use of PTZ as a carbapenem-sparing agent in ESBL-producing infections.

The study showed that ASTar can provide MIC of PTZ on the same working day as detecting an *E. coli*-positive blood culture, with strong concordance to reference, even in isolates that pose difficulties with other commercial devices. With this improvement in the accuracy of piperacillin/tazobactam sensitivity, this combination could be re-evaluated for use in ESBL-producing infections.

Looking to the future

At Q-linea, it has been an exciting year where the company has expanded its operations throughout Europe, securing multiple distribution partnerships in the UK, France, Norway and Poland, and formed a legal entity in Italy to accelerate the commercialisation of ASTar.

Q-linea is also gaining momentum in the US, having completed its FDA clinical trials, and begun expanding its commercial organisation there. Several ASTar evaluations and studies have been completed and others are ongoing in several geographies. It is evident that increasing AMR and the impact it has on the treatment of infectious diseases is a global issue causing millions of potentially avoidable deaths and driving up the cost of healthcare. To tackle challenges like AMR and improve the management of septic patients, rapid AST is becoming an essential component in every diagnostics lab, and the strong interest we have seen so far shows that ASTar is ideally fitted for this cause.

Q-linea has confidence that the continuous refinement of the ASTar system helps to future-proof healthcare and will help to ensure that antibiotics continue to be an effective treatment option for future generations. ●

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